Model Architecture: Qwen3.5-9B

Generated by sg_modelload

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1 Model Overview

Model name	Qwen3.5-9B
Transformers class	`Qwen3_5ForConditionalGeneration`
Path	`/models/sayak_da_work/Qwen3.5-9B`
Total parameters	9,409,813,744 (9.41B)
Checkpoint-only tensors (not in total)	243,290,624
Weight dtype	bfloat16
Hidden size d	4,096
Decoder layers L	32
Query heads H	16
Key/value heads H_{kv}	4
Head dim d_h	256
MLP width d_{ff}	12,288
DeltaNet q/k heads	16
DeltaNet v heads	32
DeltaNet q/k head dim	128
DeltaNet v head dim	128
DeltaNet conv kernel	4
Vocabulary V	248,320
Max context length	262,144
RMSNorm ϵ	1e-06
Tied input/output embeddings	False
Activation	SiLU
RoPE base θ	10,000,000
Vision encoder: layers	27
hidden size / heads	1152 / 16

How to read this document. Section 2 writes the whole network as one composed function. Section 3 gives readable pseudocode for the forward pass and for every distinct block type. Section 4 lists the weights of each block type *once*, with how many times it repeats. Section 5 explains in plain language what every kind of layer does. The original layer-by-layer table with every repetition written out is in Appendix A.

Notation

B, T	batch size, sequence length (tokens)
d	hidden size (4096)
V	vocabulary size (248,320)
H, H_{kv}, d_h	query heads, key/value heads, size of one head
W_*, b_*	learned weight matrix / bias of the named projection
$\odot$, σ	element-wise product, logistic sigmoid
$f^{\circ n}$	f applied n times in a row ($f \circ f \circ \cdots \circ f$)
$g \circ f$	first f, then g (composition reads right to left)
RMSNorm(x)	$\gamma \odot x / \sqrt{\text{mean}(x^2) + \epsilon}$ (learned scale γ)
LN(x)	LayerNorm: $\gamma \odot (x - \mu)/\sigma + \beta$
RoPE	rotary position embedding (rotates query/key pairs by an angle set by position)

2 The Model as a Function

Input: token ids $x \in \{0, \ldots, V-1\}^{B \times T}$. Output: next-token logits $y \in \mathbb{R}^{B \times T \times V}$. The 32 decoder layers are written with composition powers, so a block that repeats n times appears once as $\mathcal{B}^{\circ n}$.

$$h_0 = \text{Merge}\Big(\text{Embed}(x),\ \text{Proj}_{\text{img}}\big(\text{Enc}_{\text{V}}(\text{img})\big)\Big)$$

$$h_L = \Big(\big(\mathcal{B}_{\text{F}} \circ \mathcal{B}_{\text{L}}^{\circ 3}\big)^{\circ 8}\Big)(h_0)$$

$$y = W_{\text{LM}}\ \text{RMSNorm}_{\text{final}}(h_L)$$

Merge puts the projected image/audio vectors at the placeholder-token positions of the text embedding sequence. The encoders are themselves repeated stacks:

$$\text{Enc}_{\text{V}} = \text{Post} \circ \mathcal{B}_{\text{V}}^{\circ 27} \circ \text{PatchEmbed}$$

2.1 Layer pattern

Symbol	Block type	Count	Params / block	Layer indices
$\mathcal{B}_{\mathrm{L}}$	Linear-attention (Gated DeltaNet) block	24	218,407,104	0–2, 4–6, 8–10, 12–14, 16–18, 20–22, 24–26, 28–30
$\mathcal{B}_{\mathrm{F}}$	Full (global) attention block	8	209,723,904	3, 7, …, 31 (every 4th)
$\mathcal{B}_{\mathrm{V}}$	Vision encoder block	27	15,239,504	0–26

Language-model decoder (32 layers). layers 0–31: $[\mathcal{B}_{\mathrm{L}}\times 3, \mathcal{B}_{\mathrm{F}}]$ repeated $8\times$.

Vision encoder (27 layers). layers 0–26: $[\mathcal{B}_{\mathrm{V}}]$ repeated $27\times$.

2.2 Block functions

$\mathcal{B}_{\mathrm{L}}$ — **Linear-attention (Gated DeltaNet) block.**

$$
\begin{aligned}
a &= h + \mathrm{Mixer}(\mathrm{RMSNorm}_1(h)) && \text{pre-norm + residual}\\
\mathcal{B}(h) &= a + \mathrm{FFN}(\mathrm{RMSNorm}_2(a)) && \text{pre-norm + residual}\\
[q,\, k,\, v] &= \mathrm{SiLU}\left(\mathrm{CausalConv1d}(W_{qkv}\, u)\right) && \text{short causal conv (kernel 4) over time}\\
z &= W_z\, u && \text{output gate}\\
\beta_t &= \sigma(W_b\, u_t) && \text{write strength in (0,1)}\\
g_t &= -e^{A_{\log}} \cdot \mathrm{softplus}(W_a\, u_t + b_{dt}) && \text{decay } (\le 0)\text{: how fast memory fades}\\
q_t,\, k_t &= \mathrm{L2Norm}(q_t),\ \mathrm{L2Norm}(k_t)\\
S_t &= e^{g_t}\, S_{t-1}\left(I - \beta_t k_t k_t^{\top}\right) + \beta_t\, v_t k_t^{\top} && \text{fixed-size memory per head (delta rule)}\\
o_t &= S_t\, q_t && \text{read memory with the query}\\
\mathrm{Mixer}(u) &= W_{\mathrm{out}}\left(\mathrm{RMSNorm}(o) \odot \mathrm{SiLU}(z)\right)\\
\mathrm{FFN}(u) &= W_{\mathrm{down}}\left(\mathrm{SiLU}(W_{\mathrm{gate}}\, u) \odot W_{\mathrm{up}}\, u\right) && \text{gated MLP, width 12288}
\end{aligned}
$$

$\mathcal{B}_{\mathrm{F}}$ — **Full (global) attention block.**

$$
\begin{aligned}
a &= h + \mathrm{Attn}(\mathrm{RMSNorm}_1(h)) && \text{pre-norm + residual}\\
\mathcal{B}(h) &= a + \mathrm{FFN}(\mathrm{RMSNorm}_2(a)) && \text{pre-norm + residual}\\
[Q,\, G] &= W_Q\, u && \text{query and an output gate}\\
K,\, V &= W_K\, u, \quad W_V\, u && H_{kv} = 4 \text{ key/value heads}\\
Q,\, K &= \mathrm{RMSNorm}_q(Q),\ \mathrm{RMSNorm}_k(K) && \text{per-head normalisation}\\
Q,\, K &= \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K) && \text{rotate by position}\\
S_h &= s\, Q_h K_{g(h)}^{\top} + M && \text{mask } M\text{: causal}\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\, V_{g(h)}\\
\mathrm{Attn}(u) &= W_O\left([O_1; \ldots; O_H] \odot \sigma(G)\right)\\
\mathrm{FFN}(u) &= W_{\mathrm{down}}\left(\mathrm{SiLU}(W_{\mathrm{gate}}\, u) \odot W_{\mathrm{up}}\, u\right) && \text{gated MLP, width 12288}
\end{aligned}
$$

$\mathcal{B}_{\mathrm{V}}$ — **Vision encoder block.**

$$
\begin{aligned}
a &= h + \mathrm{Attn}(\mathrm{LN}_1(h)) && \text{pre-norm + residual}\\
\mathcal{B}(h) &= a + \mathrm{FFN}(\mathrm{LN}_2(a)) && \text{pre-norm + residual}\\
[Q,\, K,\, V] &= W_{qkv}\, u + b_{qkv} && \text{one fused projection}\\
Q,\, K &= \mathrm{RoPE}_{2D}(Q),\ \mathrm{RoPE}_{2D}(K) && \text{rotate by (row, column) patch position}\\
S_h &= s\, Q_h K_h^{\top} + M && \text{mask } M\text{: bidirectional}\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\, V_h\\
\mathrm{Attn}(u) &= W_O\, [O_1; \ldots; O_H] + b_O\\
\mathrm{FFN}(u) &= W_2\ \mathrm{GELU}_{\tanh}(W_1 u + b_1) + b_2 && \text{MLP, width 4304}
\end{aligned}
$$

3 Forward Pass in Pseudocode

Algorithm 1 Qwen3.5-9B: full forward pass

Require: token ids x of shape $B \times T$; optional images / audio

$h \leftarrow \text{Embed}(x)$ ▷ look up one vector per token: $B{\times}T{\times}4096$
if image input present **then**
  $e \leftarrow \text{Enc}_{\text{V}}(\text{image})$ ▷ vision encoder: 27 layers, Algorithm 2
  $h[\text{image token slots}] \leftarrow \text{Proj}(e)$ ▷ project into text width 4096
end if
for $r \leftarrow 0$ **to** 7 **do** ▷ layers 0–31: pattern of 4 repeated 8 times
  for $j \leftarrow 1$ **to** 3 **do**
    $h \leftarrow \mathcal{B}_{\text{L}}(h)$ ▷ Linear-attention (Gated DeltaNet) block
  end for
  $h \leftarrow \mathcal{B}_{\text{F}}(h)$ ▷ Full (global) attention block
end for
$h \leftarrow \text{RMSNorm}_{\text{final}}(h)$
$y \leftarrow h\, W_{\text{LM}}^{\top}$ ▷ logits $B{\times}T{\times}248,320$
return y ▷ softmax(y) = next-token probabilities

Algorithm 2 Vision encoder

$h \leftarrow \text{PatchEmbed}(\text{pixels}) + \text{position embedding}$ ▷ one vector per image patch
for $\ell \leftarrow 0$ **to** 26 **do**
  $h \leftarrow \mathcal{B}_{\text{V}}(h)$
end for
$h \leftarrow \text{Post}(h)$ ▷ final norm / pooling / merging, see Table 2
return h

Algorithm 3 Block $\mathcal{B}_{\text{L}}(h)$ — Linear-attention (Gated DeltaNet) block

Require: h of shape $B \times T \times 4096$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \text{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: Gated DeltaNet (linear attention, fixed-size memory)
$q, k, v \leftarrow \text{split}(\text{SiLU}(\text{CausalConv1d}(uW_{qkv}^{\top})))$ ▷ 16 q/k heads of 128, 32 v heads of 128
$z \leftarrow uW_{z}^{\top};\ \beta \leftarrow \sigma(uW_{b}^{\top});\ g \leftarrow -e^{A_{\log}}\,\text{softplus}(uW_{a}^{\top} + b_{dt})$ ▷ gate, write strength, decay
for $t \leftarrow 1$ **to** T **do** ▷ conceptually sequential; computed in chunks
  $S \leftarrow e^{g_t} S\,(I - \beta_t k_t k_t^{\top}) + \beta_t v_t k_t^{\top}$ ▷ forget a bit, overwrite what k_t points at
  $o_t \leftarrow S\, q_t$ ▷ read
end for
$m \leftarrow \text{RMSNorm}(o) \odot \text{SiLU}(z)\ W_{\text{out}}^{\top}$ ▷ back to $B{\times}T{\times}4096$
$h \leftarrow r + m$ ▷ add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \text{RMSNorm}(h)$
$m \leftarrow \big(\text{SiLU}(uW_{\text{gate}}^{\top}) \odot uW_{\text{up}}^{\top}\big) W_{\text{down}}^{\top}$ ▷ expand to 12288, gate, shrink back to 4096
$h \leftarrow r + m$ ▷ add residual
return h

Algorithm 4 Block $\mathcal{B}_{\mathrm{F}}(h)$ — Full (global) attention block

Require: h of shape $B \times T \times 4096$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^\top$ ▷ $B{\times}T{\times}16{\times}256$ (16 query heads); second half is a gate G
$K \leftarrow uW_K^\top;\ V \leftarrow uW_V^\top$ ▷ $B{\times}T{\times}4{\times}256$ (4 key/value heads, each shared by 4 query heads)
$Q \leftarrow \mathrm{RMSNorm}(Q);\ K \leftarrow \mathrm{RMSNorm}(K)$ ▷ normalise each head vector (stabilises training)
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs
$P \leftarrow \mathrm{softmax}(s\,QK^\top + M)$ ▷ mask: causal: a token sees only earlier tokens
$O \leftarrow PV$ ▷ $B{\times}T{\times}16{\times}256$
$O \leftarrow O \odot \sigma(G)$ ▷ output gate decides how much of each head to keep
$m \leftarrow O\,W_O^\top$ ▷ concatenate heads, project back to $B{\times}T{\times}4096$
$h \leftarrow r + m$ ▷ add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$m \leftarrow \big(\mathrm{SiLU}(uW_{\mathrm{gate}}^\top) \odot uW_{\mathrm{up}}^\top\big)W_{\mathrm{down}}^\top$ ▷ expand to 12288, gate, shrink back to 4096
$h \leftarrow r + m$ ▷ add residual
return h

Algorithm 5 Block $\mathcal{B}_{\mathrm{V}}(h)$ — Vision encoder block

Require: h of shape $B \times T \times 1152$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{LN}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q, K, V \leftarrow \mathrm{split}(uW_{qkv}^\top + b)$ ▷ each $B{\times}T{\times}16{\times}72$
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ 2-D rotary position (row, column)
$P \leftarrow \mathrm{softmax}(s\,QK^\top + M)$ ▷ mask: every patch/frame sees every other
$O \leftarrow PV$ ▷ $B{\times}T{\times}16{\times}72$
$m \leftarrow O\,W_O^\top + b_O$ ▷ concatenate heads, project back to $B{\times}T{\times}1152$
$h \leftarrow r + m$ ▷ add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{LN}(h)$
$m \leftarrow \mathrm{GELU}_{\tanh}(uW_1^\top + b_1)W_2^\top + b_2$ ▷ expand to 4304, shrink back to 1152
$h \leftarrow r + m$ ▷ add residual
return h

4 Compact Architecture (repeated layers written once)

4.1 Modules outside the repeated stacks

Table 2: Weights that appear once

Weight	Type	Shape	Params
`model.visual.patch_embed.proj.weight`	Conv3d	$1152 \times 3 \times 2 \times 16 \times 16$	1,769,472
`model.visual.patch_embed.proj.bias`	Conv3d	1152	1,152
`model.visual.pos_embed.weight`	Embedding	2304×1152	2,654,208
`model.visual.merger.norm.weight`	LayerNorm	1152	1,152
`model.visual.merger.norm.bias`	LayerNorm	1152	1,152
`model.visual.merger.linear_fc1.weight`	Linear	4608×4608	21,233,664
`model.visual.merger.linear_fc1.bias`	Linear	4608	4,608
`model.visual.merger.linear_fc2.weight`	Linear	4096×4608	18,874,368
`model.visual.merger.linear_fc2.bias`	Linear	4096	4,096
`model.language_model.embed_tokens.weight`	Embedding	248320×4096	1,017,118,720
`model.language_model.norm.weight`	Qwen3_5RMSNorm	4096	4,096
`lm_head.weight`	Linear	248320×4096	1,017,118,720

4.2 Language-model decoder: `model.language_model.layers`

Table 3: $\mathcal{B}_{\mathrm{L}}$: Linear-attention (Gated DeltaNet) block — layers 0–2, 4–6, 8–10, 12–14, 16–18, 20–22, 24–26, 28–30; weight names are relative to `model.language_model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`linear_attn.dt_bias`	Qwen3_5GatedDeltaNet	32	32
`linear_attn.A_log`	Qwen3_5GatedDeltaNet	32	32
`linear_attn.conv1d.weight`	Conv1d	$8192 \times 1 \times 4$	32,768
`linear_attn.norm.weight`	Qwen3_5RMSNormGated	128	128
`linear_attn.out_proj.weight`	Linear	4096×4096	16,777,216
`linear_attn.in_proj_qkv.weight`	Linear	8192×4096	33,554,432
`linear_attn.in_proj_z.weight`	Linear	4096×4096	16,777,216
`linear_attn.in_proj_b.weight`	Linear	32×4096	131,072
`linear_attn.in_proj_a.weight`	Linear	32×4096	131,072
`mlp.gate_proj.weight`	Linear	12288×4096	50,331,648
`mlp.up_proj.weight`	Linear	12288×4096	50,331,648
`mlp.down_proj.weight`	Linear	4096×12288	50,331,648
`input_layernorm.weight`	Qwen3_5RMSNorm	4096	4,096
`post_attention_layernorm.weight`	Qwen3_5RMSNorm	4096	4,096
Per block			**218,407,104**
× 24 blocks			**5,241,770,496**

Table 4: $\mathcal{B}_{\mathrm{F}}$: Full (global) attention block — layers 3, 7, …, 31 (every 4th); weight names are relative to `model.language_model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.q_proj.weight`	Linear	8192×4096	33,554,432
`self_attn.k_proj.weight`	Linear	1024×4096	4,194,304
`self_attn.v_proj.weight`	Linear	1024×4096	4,194,304
`self_attn.o_proj.weight`	Linear	4096×4096	16,777,216
`self_attn.q_norm.weight`	Qwen3_5RMSNorm	256	256
`self_attn.k_norm.weight`	Qwen3_5RMSNorm	256	256
`mlp.gate_proj.weight`	Linear	12288×4096	50,331,648
`mlp.up_proj.weight`	Linear	12288×4096	50,331,648
`mlp.down_proj.weight`	Linear	4096×12288	50,331,648
`input_layernorm.weight`	Qwen3_5RMSNorm	4096	4,096
`post_attention_layernorm.weight`	Qwen3_5RMSNorm	4096	4,096
Per block			**209,723,904**
× 8 blocks			**1,677,791,232**

4.3 Vision encoder: `model.visual.blocks`

Table 5: $\mathcal{B}_{\mathrm{V}}$: Vision encoder block — layers 0–26; weight names are relative to `model.visual.blocks.`ℓ

Weight (relative)	Type	Shape	Params
`norm1.weight`	LayerNorm	1152	1,152
`norm1.bias`	LayerNorm	1152	1,152
`norm2.weight`	LayerNorm	1152	1,152
`norm2.bias`	LayerNorm	1152	1,152
`attn.qkv.weight`	Linear	3456×1152	3,981,312

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Weight (relative)	Type	Shape	Params
`attn.qkv.bias`	Linear	3456	3,456
`attn.proj.weight`	Linear	1152×1152	1,327,104
`attn.proj.bias`	Linear	1152	1,152
`mlp.linear_fc1.weight`	Linear	4304×1152	4,958,208
`mlp.linear_fc1.bias`	Linear	4304	4,304
`mlp.linear_fc2.weight`	Linear	1152×4304	4,958,208
`mlp.linear_fc2.bias`	Linear	1152	1,152
Per block			**15,239,504**
× 27 blocks			**411,466,608**

4.4 Checkpoint-only modules

Multi-token-prediction (MTP) head (`mtp.*`). An extra one-layer transformer that predicts the token after next. It concatenates RMSNorm(embedding of token $t{+}1$) with RMSNorm(final hidden state at t), projects $2d \to d$ (`fc`), runs one full-attention decoder layer, then a norm and the shared LM head. Serving engines (vLLM/SGLang) use it for speculative decoding. The transformers class does not instantiate it, so it is not counted in the total above.

Tensor	Shape	Dtype	Params
`mtp.fc.weight`	4096×8192	bf16	33,554,432
`mtp.layers.0.input_layernorm.weight`	4096	bf16	4,096
`mtp.layers.0.mlp.down_proj.weight`	4096×12288	bf16	50,331,648
`mtp.layers.0.mlp.gate_proj.weight`	12288×4096	bf16	50,331,648
`mtp.layers.0.mlp.up_proj.weight`	12288×4096	bf16	50,331,648
`mtp.layers.0.post_attention_layernorm.weight`	4096	bf16	4,096
`mtp.layers.0.self_attn.k_norm.weight`	256	bf16	256
`mtp.layers.0.self_attn.k_proj.weight`	1024×4096	bf16	4,194,304
`mtp.layers.0.self_attn.o_proj.weight`	4096×4096	bf16	16,777,216
`mtp.layers.0.self_attn.q_norm.weight`	256	bf16	256
`mtp.layers.0.self_attn.q_proj.weight`	8192×4096	bf16	33,554,432
`mtp.layers.0.self_attn.v_proj.weight`	1024×4096	bf16	4,194,304
`mtp.norm.weight`	4096	bf16	4,096
`mtp.pre_fc_norm_embedding.weight`	4096	bf16	4,096
`mtp.pre_fc_norm_hidden.weight`	4096	bf16	4,096

5 What Each Layer Does

One row per *kind* of layer present in this model. "Appears as" gives the module names (layer indices removed).

Layer (appears as)	What it does
LANGUAGE MODEL	
Token embedding `embed_tokens`	Lookup table with one learned vector per vocabulary token. Turns token ids into the vectors the network computes with (Gemma multiplies them by $\sqrt{d}$).
Query projection (q_proj) `self_attn.q_proj`	Turns each token vector into query vectors, one per head: "what am I looking for?". Here it also outputs an equally sized gate for the attention output (hence the doubled width).
Query norm (q_norm) `self_attn.q_norm`	RMSNorm applied to every query head vector before RoPE. Keeps attention logits in a sane range (QK-norm).
Key projection (k_proj) `self_attn.k_proj`	Turns each token into key vectors: "what do I contain?". Queries are compared with keys to decide where to attend. With grouped-query attention there are fewer key heads than query heads.
Key norm (k_norm) `self_attn.k_norm`	RMSNorm applied to every key head vector before RoPE.
Value projection (v_proj) `self_attn.v_proj`	Turns each token into value vectors: the information actually copied to tokens that attend to it.

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Layer (appears as)	What it does
DeltaNet output projection `linear_attn.out_proj`	Mixes the heads back into one hidden-size vector.
Output projection (o_proj) `self_attn.o_proj`	Concatenates all heads' results and mixes them back into one hidden-size vector per token.
LM head (output projection) `lm_head`	Maps each final hidden vector to one score (logit) per vocabulary token. Softmax of the logits gives the next-token probabilities. When tied, it reuses the embedding matrix.
DeltaNet Q/K/V projection `linear_attn.in_proj_qkv`	Produces queries, keys and values for the linear-attention layer in one matrix.
DeltaNet output-gate projection `linear_attn.in_proj_z`	Produces gate z. The normalised output is multiplied by SiLU(z).
DeltaNet write-strength projection `linear_attn.in_proj_b`	Produces $\beta = \sigma(\cdot)$ per head: how strongly the current token overwrites the memory.
DeltaNet decay projection `linear_attn.in_proj_a`	Produces the input-dependent forgetting rate (with A_log and dt_bias).
Short causal convolution `linear_attn.conv1d`	Depthwise 1-D convolution over the last 4 tokens of Q/K/V. Gives cheap local context before the recurrent memory update.
A_log (decay base) `layers.linear_attn`	Per-head learned log decay rate. $-e^{A_{\log}}$ sets how quickly old memory fades.
dt_bias (decay bias) `layers.linear_attn`	Per-head bias added before the softplus that controls the decay step.
Gated output RMSNorm `linear_attn.norm`	Normalises each head's memory read-out and multiplies it by SiLU(z).
Gate projection (gate_proj) `mlp.gate_proj`	Expands the token vector to the MLP width and passes it through SiLU/GELU. The result acts as a soft on/off switch for each hidden unit.
Up projection (up_proj) `mlp.up_proj`	Expands the token vector to the MLP width (the "content" path). It is multiplied element-wise by the gate (SwiGLU/GeGLU).
Down projection (down_proj) `mlp.down_proj`	Shrinks the gated hidden vector back to the model width so it can be added to the residual stream.
Input norm (input_layernorm) `layers.input_layernorm`	RMSNorm applied to the residual stream before attention. Rescales each token vector to unit RMS, then multiplies by a learned per-channel weight.
post_attention_layernorm `layers.post_attention_layernorm`	Despite the name, this is the pre-MLP norm: RMSNorm on the residual stream after attention, feeding the MLP.
Final norm `norm`	RMSNorm applied once after the last decoder layer, just before the LM head.
VISION ENCODER	
3-D patch embedding (Conv3d) `patch_embed.proj`	Cuts images/videos into 2-frame × 16 × 16-pixel cubes and maps each cube to one vector.
Position embedding `visual.pos_embed`	Learned position vectors, interpolated to the image's patch grid.
Pre-attention LayerNorm `blocks.norm1`	LayerNorm before vision self-attention.
Pre-MLP LayerNorm `blocks.norm2`	LayerNorm before the vision MLP.
Fused Q/K/V projection `attn.qkv`	One matrix that produces queries, keys and values at once (split afterwards).
Attention output projection `attn.proj`	Mixes the concatenated head outputs back to the model width.
MLP up-projection (fc1) `mlp.linear_fc1`	Expands each vector to the wider MLP size before the GELU non-linearity.
MLP down-projection (fc2) `mlp.linear_fc2`	Shrinks back to the model width.

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Layer (appears as)	What it does
Merger LayerNorm `merger.norm`	Normalises patch features before 2×2 neighbouring patches are merged.
Merger MLP layer 1 `merger.linear_fc1`	Takes 4 neighbouring patches concatenated (2×2 merge, so 4× fewer image tokens) and mixes them.
Merger MLP layer 2 (to text width) `merger.linear_fc2`	Projects merged patches into the language model's hidden size.

6 Parameter Count

Table 8: Parameters by component

Component	Parameters	% of total
MLP / MoE	4,831,838,208	51.3%
Attention / token mixer	2,087,461,376	22.2%
Token embedding	1,017,118,720	10.8%
LM head	1,017,118,720	10.8%
Vision encoder	456,010,480	4.8%
Norms	266,240	0.0%
Total	**9,409,813,744**	**100.0%**

Table 9: Parameters by block type

Block type	Count	Params / block	Total
$\mathcal{B}_{\mathrm{L}}$ Linear-attention (Gated DeltaNet) block	24	218,407,104	5,241,770,496
$\mathcal{B}_{\mathrm{F}}$ Full (global) attention block	8	209,723,904	1,677,791,232
$\mathcal{B}_{\mathrm{V}}$ Vision encoder block	27	15,239,504	411,466,608
Modules outside stacks	1	–	2,078,785,408
Total			**9,409,813,744**

Appendix A Full Layer-by-Layer Table

Every weight tensor with every repetition written out (the original table).

Layer path	Type	Shape	Dtype	Params
`lm_head.weight`	Linear	248320 × 4096	bfloat16	1,017,118,720
`model`	Qwen3_5Model	–	–	–
`model.language_model`	Qwen3_5TextModel	–	–	–
`model.language_model.embed_tokens.weight`	Embedding	248320 × 4096	bfloat16	1,017,118,720
`model.language_model.layers.0`	Qwen3_5DecoderLayer	–	–	–
`model.language_model.layers.0.input_layernorm.weight`	Qwen3_5RMSNorm	4096	bfloat16	4,096
`model.language_model.layers.0.linear_attn.A_log`	Qwen3_5GatedDeltaNet	32	bfloat16	32
`model.language_model.layers.0.linear_attn`	Qwen3_5GatedDeltaNet	–	–	–
`model.language_model.layers.0.linear_attn.conv1d.weight`	Conv1d	8192 × 1 × 4	bfloat16	32,768
`model.language_model.layers.0.linear_attn.dt_bias`	Qwen3_5GatedDeltaNet	32	bfloat16	32
`model.language_model.layers.0.linear_attn.in_proj_a.weight`	Linear	32 × 4096	bfloat16	131,072
`model.language_model.layers.0.linear_attn.in_proj_b.weight`	Linear	32 × 4096	bfloat16	131,072
`model.language_model.layers.0.linear_attn.in_proj_qkv.weight`	Linear	8192 × 4096	bfloat16	33,554,432
`model.language_model.layers.0.linear_attn.in_proj_z.weight`	Linear	4096 × 4096	bfloat16	16,777,216
`model.language_model.layers.0.linear_attn.norm.weight`	Qwen3_5RMSNorm Gated	128	bfloat16	128

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.0.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.0.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.0.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.0.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.0.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.0.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.1	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.1.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.1.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.1.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.1.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.1.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.1.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.1.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.1.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.1.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.1.linear_attn.norm.weight	Qwen3_5RMSNorm Gated	128	bfloat16	128
model.language_model.layers.1.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.1.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.1.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.1.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.1.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.1.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.2	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.2.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.2.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.2.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.2.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.2.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.2.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.2.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.2.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.2.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.2.linear_attn.norm.weight	Qwen3_5RMSNorm Gated	128	bfloat16	128
model.language_model.layers.2.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.2.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.2.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.2.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.2.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.2.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.3	Qwen3_5DecoderLayer	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.3.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.3.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.3.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.3.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.3.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.3.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.3.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.3.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.3.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.3.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.3.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.3.self_attn.q_proj.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.3.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.4	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.4.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.4.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.4.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.4.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.4.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.4.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.4.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.4.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.4.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.4.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.4.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.4.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.4.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.4.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.4.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.4.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.5	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.5.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.5.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.5.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.5.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.5.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.5.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.5.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.5.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.5.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.5.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.5.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.5.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.5.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.5.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.5.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.5.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.6	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.6.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.6.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.6.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.6.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.6.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.6.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.6.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.6.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.6.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.6.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.6.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.6.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.6.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.6.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.6.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.6.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.7	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.7.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.7.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.7.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.7.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.7.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.7.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.7.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.7.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.7.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.7.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.7.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.7.self_attn.q_proj.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.7.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.8	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.8.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.8.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.8.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.8.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.8.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.8.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.8.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.8.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.8.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.8.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.8.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.8.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.8.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.8.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.8.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.8.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.9	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.9.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.9.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.9.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.9.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.9.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.9.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.9.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.9.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.9.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.9.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.9.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.9.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.9.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.9.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.9.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.9.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.10	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.10.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.10.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.10.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.10.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.10.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.10.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.10.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.10.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.10.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.10.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.10.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.10.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.10.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.10.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.10.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.10.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.11	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.11.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.11.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.11.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.11.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.11.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.11.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.11.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.11.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.11.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.11.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.11.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.11.self_attn.q_proj.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.11.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.12	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.12.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.12.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.12.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.12.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.12.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.12.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.12.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.12.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.12.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.12.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.12.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.12.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.12.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.12.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.12.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.12.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.13	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.13.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.13.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.13.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.13.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.13.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.13.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.13.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.13.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.13.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.13.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.13.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.13.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.13.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.13.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.13.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.13.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.14	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.14.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.14.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.14.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.14.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.14.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.14.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.14.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.14.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.14.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.14.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.14.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.14.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.14.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.14.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.14.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.14.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.15	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.15.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.15.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.15.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.15.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.15.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.15.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.15.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.15.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.15.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.15.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.15.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.15.self_attn.q_proj.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.15.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.16	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.16.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.16.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.16.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.16.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.16.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.16.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.16.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.16.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.16.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.16.linear_attn.norm.weight	Qwen3_5RMSNorm Gated	128	bfloat16	128
model.language_model.layers.16.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.16.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.16.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.16.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.16.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.16.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.17	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.17.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.17.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.17.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.17.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.17.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.17.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.17.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.17.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.17.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.17.linear_attn.norm.weight	Qwen3_5RMSNorm Gated	128	bfloat16	128
model.language_model.layers.17.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.17.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.17.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.17.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.17.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.17.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.18	Qwen3_5DecoderLayer	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.18.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.18.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.18.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.18.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.18.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.18.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.18.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.18.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.18.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.18.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.18.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.18.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.18.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.18.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.18.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.18.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.19	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.19.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.19.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.19.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.19.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.19.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.19.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.19.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.19.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.19.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.19.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.19.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.19.self_attn.q_proj.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.19.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.20	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.20.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.20.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.20.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.20.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.20.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.20.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.20.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.20.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.20.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.20.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.20.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.20.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.20.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.20.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.20.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.20.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.21	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.21.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.21.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.21.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.21.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.21.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.21.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.21.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.21.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.21.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.21.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.21.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.21.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.21.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.21.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.21.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.21.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.22	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.22.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.22.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.22.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.22.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.22.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.22.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.22.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.22.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.22.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.22.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.22.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.22.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.22.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.22.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.22.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.22.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.23	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.23.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.23.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.23.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.23.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.23.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.23.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.23.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.23.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.23.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.23.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.23.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.23.self_attn.q_proj.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.23.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.24	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.24.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.24.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.24.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.24.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.24.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.24.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.24.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.24.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.24.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.24.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.24.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.24.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.24.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.24.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.24.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.24.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.25	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.25.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.25.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.25.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.25.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.25.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.25.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.25.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.25.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.25.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.25.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.25.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.25.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.25.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.25.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.25.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.25.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.26	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.26.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.26.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.26.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.26.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.26.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.26.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.26.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.26.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.26.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.26.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.26.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.26.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.26.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.26.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.26.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.26.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.27	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.27.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.27.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.27.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.27.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.27.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.27.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.27.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.27.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.27.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.27.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.27.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.27.self_attn.q_proj.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.27.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.28	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.28.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.28.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.28.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.28.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.28.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.28.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.28.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.28.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.28.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.28.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.28.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.28.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.28.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.28.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.28.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.28.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.29	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.29.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.29.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.29.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.29.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.29.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.29.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.29.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.29.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.29.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.29.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.29.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.29.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.29.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.29.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.29.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.29.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.30	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.30.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.30.linear_attn.A_log	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.30.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.30.linear_attn.conv1d.weight	Conv1d	8192 × 1 × 4	bfloat16	32,768
model.language_model.layers.30.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	32	bfloat16	32
model.language_model.layers.30.linear_attn.in_proj_a.weight	Linear	32 × 4096	bfloat16	131,072
model.language_model.layers.30.linear_attn.in_proj_b.weight	Linear	32 × 4096	bfloat16	131,072

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.30.linear_attn.in_proj_qkv.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.30.linear_attn.in_proj_z.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.30.linear_attn.norm.weight	Qwen3_5RMSNorm Gated	128	bfloat16	128
model.language_model.layers.30.linear_attn.out_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.30.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.30.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.30.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.30.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.30.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.31	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.31.input_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.31.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.31.mlp.down_proj.weight	Linear	4096 × 12288	bfloat16	50,331,648
model.language_model.layers.31.mlp.gate_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.31.mlp.up_proj.weight	Linear	12288 × 4096	bfloat16	50,331,648
model.language_model.layers.31.post_attention_layernorm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.language_model.layers.31.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.31.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.31.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.layers.31.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.language_model.layers.31.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.31.self_attn.q_proj.weight	Linear	8192 × 4096	bfloat16	33,554,432
model.language_model.layers.31.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.language_model.norm.weight	Qwen3_5RMSNorm	4096	bfloat16	4,096
model.visual	Qwen3_5VisionModel	–	–	–
model.visual.blocks.0	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.0.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.0.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.0.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.0.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.0.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.0.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.0.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.0.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.0.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.0.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.0.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.0.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.0.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.0.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.1	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.1.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.1.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.1.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.1.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.1.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.1.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.1.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.1.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.1.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.1.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.1.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.1.norm1.weight	LayerNorm	1152	bfloat16	1,152

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Layer path	Type	Shape	Dtype	Params
model.visual.blocks.1.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.1.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.2	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.2.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.2.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.2.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.2.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.2.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.2.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.2.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.2.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.2.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.2.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.2.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.2.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.2.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.2.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.3	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.3.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.3.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.3.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.3.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.3.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.3.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.3.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.3.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.3.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.3.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.3.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.3.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.3.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.3.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.4	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.4.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.4.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.4.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.4.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.4.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.4.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.4.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.4.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.4.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.4.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.4.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.4.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.4.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.4.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.5	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.5.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.5.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.5.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.5.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.5.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.5.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.5.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.5.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.5.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.5.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.5.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.5.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.5.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.5.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.6	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.6.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.6.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.6.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.6.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.6.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.6.mlp	Qwen3_5VisionMLP	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.visual.blocks.6.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.6.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.6.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.6.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.6.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.6.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.6.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.6.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.7	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.7.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.7.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.7.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.7.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.7.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.7.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.7.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.7.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.7.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.7.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.7.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.7.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.7.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.7.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.8	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.8.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.8.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.8.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.8.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.8.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.8.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.8.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.8.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.8.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.8.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.8.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.8.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.8.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.8.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.9	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.9.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.9.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.9.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.9.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.9.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.9.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.9.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.9.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.9.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.9.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.9.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.9.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.9.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.9.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.10	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.10.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.10.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.10.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.10.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.10.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.10.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.10.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.10.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.10.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.10.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.10.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.10.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.10.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.10.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.11	Qwen3_5VisionBlock	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.visual.blocks.11.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.11.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.11.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.11.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.11.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.11.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.11.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.11.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.11.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.11.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.11.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.11.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.11.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.11.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.12	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.12.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.12.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.12.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.12.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.12.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.12.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.12.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.12.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.12.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.12.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.12.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.12.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.12.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.12.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.13	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.13.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.13.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.13.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.13.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.13.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.13.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.13.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.13.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.13.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.13.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.13.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.13.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.13.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.13.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.14	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.14.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.14.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.14.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.14.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.14.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.14.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.14.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.14.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.14.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.14.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.14.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.14.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.14.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.14.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.15	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.15.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.15.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.15.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.15.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.15.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.15.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.15.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.15.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.15.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152

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Layer path	Type	Shape	Dtype	Params
model.visual.blocks.15.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.15.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.15.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.15.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.15.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.16	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.16.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.16.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.16.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.16.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.16.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.16.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.16.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.16.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.16.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.16.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.16.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.16.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.16.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.16.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.17	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.17.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.17.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.17.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.17.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.17.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.17.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.17.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.17.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.17.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.17.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.17.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.17.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.17.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.17.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.18	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.18.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.18.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.18.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.18.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.18.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.18.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.18.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.18.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.18.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.18.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.18.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.18.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.18.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.18.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.19	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.19.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.19.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.19.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.19.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.19.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.19.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.19.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.19.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.19.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.19.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.19.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.19.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.19.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.19.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.20	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.20.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.20.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.20.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104

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Layer path	Type	Shape	Dtype	Params
model.visual.blocks.20.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.20.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.20.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.20.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.20.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.20.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.20.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.20.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.20.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.20.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.20.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.21	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.21.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.21.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.21.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.21.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.21.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.21.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.21.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.21.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.21.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.21.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.21.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.21.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.21.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.21.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.22	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.22.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.22.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.22.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.22.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.22.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.22.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.22.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.22.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.22.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.22.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.22.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.22.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.22.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.22.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.23	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.23.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.23.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.23.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.23.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.23.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.23.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.23.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.23.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.23.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.23.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.23.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.23.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.23.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.23.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.24	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.24.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.24.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.24.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.24.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.24.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.24.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.24.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.24.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.24.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.24.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.24.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.24.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.24.norm2.bias	LayerNorm	1152	bfloat16	1,152

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Layer path	Type	Shape	Dtype	Params
model.visual.blocks.24.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.25	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.25.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.25.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.25.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.25.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.25.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.25.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.25.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.25.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.25.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.25.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.25.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.25.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.25.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.25.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.26	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.26.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.26.attn.proj.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.26.attn.proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.visual.blocks.26.attn.qkv.bias	Linear	3456	bfloat16	3,456
model.visual.blocks.26.attn.qkv.weight	Linear	3456 × 1152	bfloat16	3,981,312
model.visual.blocks.26.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.26.mlp.linear_fc1.bias	Linear	4304	bfloat16	4,304
model.visual.blocks.26.mlp.linear_fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.visual.blocks.26.mlp.linear_fc2.bias	Linear	1152	bfloat16	1,152
model.visual.blocks.26.mlp.linear_fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.visual.blocks.26.norm1.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.26.norm1.weight	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.26.norm2.bias	LayerNorm	1152	bfloat16	1,152
model.visual.blocks.26.norm2.weight	LayerNorm	1152	bfloat16	1,152
model.visual.merger	Qwen3_5VisionPatch Merger	–	–	–
model.visual.merger.linear_fc1.bias	Linear	4608	bfloat16	4,608
model.visual.merger.linear_fc1.weight	Linear	4608 × 4608	bfloat16	21,233,664
model.visual.merger.linear_fc2.bias	Linear	4096	bfloat16	4,096
model.visual.merger.linear_fc2.weight	Linear	4096 × 4608	bfloat16	18,874,368
model.visual.merger.norm.bias	LayerNorm	1152	bfloat16	1,152
model.visual.merger.norm.weight	LayerNorm	1152	bfloat16	1,152
model.visual.patch_embed	Qwen3_5VisionPatch Embed	–	–	–
model.visual.patch_embed.proj.bias	Conv3d	1152	bfloat16	1,152
model.visual.patch_embed.proj.weight	Conv3d	1152 × 3 × 2 × 16 × 16	bfloat16	1,769,472
model.visual.pos_embed.weight	Embedding	2304 × 1152	bfloat16	2,654,208

Appendix B Per-Layer Parameter Count

Module	Params	Running total
lm_head	1,017,118,720	1,017,118,720
model.language_model.embed_tokens	1,017,118,720	2,034,237,440
model.language_model.layers.0.input_layernorm	4,096	2,034,241,536
model.language_model.layers.0.linear_attn	64	2,034,241,600
model.language_model.layers.0.linear_attn.conv1d	32,768	2,034,274,368
model.language_model.layers.0.linear_attn.in_proj_a	131,072	2,034,405,440
model.language_model.layers.0.linear_attn.in_proj_b	131,072	2,034,536,512
model.language_model.layers.0.linear_attn. in_proj_qkv	33,554,432	2,068,090,944
model.language_model.layers.0.linear_attn.in_proj_z	16,777,216	2,084,868,160
model.language_model.layers.0.linear_attn.norm	128	2,084,868,288
model.language_model.layers.0.linear_attn.out_proj	16,777,216	2,101,645,504
model.language_model.layers.0.mlp.down_proj	50,331,648	2,151,977,152
model.language_model.layers.0.mlp.gate_proj	50,331,648	2,202,308,800
model.language_model.layers.0.mlp.up_proj	50,331,648	2,252,640,448

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Module	Params	Running total
model.language_model.layers.0. post_attention_layernorm	4,096	2,252,644,544
model.language_model.layers.1.input_layernorm	4,096	2,252,648,640
model.language_model.layers.1.linear_attn	64	2,252,648,704
model.language_model.layers.1.linear_attn.conv1d	32,768	2,252,681,472
model.language_model.layers.1.linear_attn.in_proj_a	131,072	2,252,812,544
model.language_model.layers.1.linear_attn.in_proj_b	131,072	2,252,943,616
model.language_model.layers.1.linear_attn. in_proj_qkv	33,554,432	2,286,498,048
model.language_model.layers.1.linear_attn.in_proj_z	16,777,216	2,303,275,264
model.language_model.layers.1.linear_attn.norm	128	2,303,275,392
model.language_model.layers.1.linear_attn.out_proj	16,777,216	2,320,052,608
model.language_model.layers.1.mlp.down_proj	50,331,648	2,370,384,256
model.language_model.layers.1.mlp.gate_proj	50,331,648	2,420,715,904
model.language_model.layers.1.mlp.up_proj	50,331,648	2,471,047,552
model.language_model.layers.1. post_attention_layernorm	4,096	2,471,051,648
model.language_model.layers.2.input_layernorm	4,096	2,471,055,744
model.language_model.layers.2.linear_attn	64	2,471,055,808
model.language_model.layers.2.linear_attn.conv1d	32,768	2,471,088,576
model.language_model.layers.2.linear_attn.in_proj_a	131,072	2,471,219,648
model.language_model.layers.2.linear_attn.in_proj_b	131,072	2,471,350,720
model.language_model.layers.2.linear_attn. in_proj_qkv	33,554,432	2,504,905,152
model.language_model.layers.2.linear_attn.in_proj_z	16,777,216	2,521,682,368
model.language_model.layers.2.linear_attn.norm	128	2,521,682,496
model.language_model.layers.2.linear_attn.out_proj	16,777,216	2,538,459,712
model.language_model.layers.2.mlp.down_proj	50,331,648	2,588,791,360
model.language_model.layers.2.mlp.gate_proj	50,331,648	2,639,123,008
model.language_model.layers.2.mlp.up_proj	50,331,648	2,689,454,656
model.language_model.layers.2. post_attention_layernorm	4,096	2,689,458,752
model.language_model.layers.3.input_layernorm	4,096	2,689,462,848
model.language_model.layers.3.mlp.down_proj	50,331,648	2,739,794,496
model.language_model.layers.3.mlp.gate_proj	50,331,648	2,790,126,144
model.language_model.layers.3.mlp.up_proj	50,331,648	2,840,457,792
model.language_model.layers.3. post_attention_layernorm	4,096	2,840,461,888
model.language_model.layers.3.self_attn.k_norm	256	2,840,462,144
model.language_model.layers.3.self_attn.k_proj	4,194,304	2,844,656,448
model.language_model.layers.3.self_attn.o_proj	16,777,216	2,861,433,664
model.language_model.layers.3.self_attn.q_norm	256	2,861,433,920
model.language_model.layers.3.self_attn.q_proj	33,554,432	2,894,988,352
model.language_model.layers.3.self_attn.v_proj	4,194,304	2,899,182,656
model.language_model.layers.4.input_layernorm	4,096	2,899,186,752
model.language_model.layers.4.linear_attn	64	2,899,186,816
model.language_model.layers.4.linear_attn.conv1d	32,768	2,899,219,584
model.language_model.layers.4.linear_attn.in_proj_a	131,072	2,899,350,656
model.language_model.layers.4.linear_attn.in_proj_b	131,072	2,899,481,728
model.language_model.layers.4.linear_attn. in_proj_qkv	33,554,432	2,933,036,160
model.language_model.layers.4.linear_attn.in_proj_z	16,777,216	2,949,813,376
model.language_model.layers.4.linear_attn.norm	128	2,949,813,504
model.language_model.layers.4.linear_attn.out_proj	16,777,216	2,966,590,720
model.language_model.layers.4.mlp.down_proj	50,331,648	3,016,922,368
model.language_model.layers.4.mlp.gate_proj	50,331,648	3,067,254,016
model.language_model.layers.4.mlp.up_proj	50,331,648	3,117,585,664

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Module	Params	Running total
model.language_model.layers.4.post_attention_layernorm	4,096	3,117,589,760
model.language_model.layers.5.input_layernorm	4,096	3,117,593,856
model.language_model.layers.5.linear_attn	64	3,117,593,920
model.language_model.layers.5.linear_attn.conv1d	32,768	3,117,626,688
model.language_model.layers.5.linear_attn.in_proj_a	131,072	3,117,757,760
model.language_model.layers.5.linear_attn.in_proj_b	131,072	3,117,888,832
model.language_model.layers.5.linear_attn.in_proj_qkv	33,554,432	3,151,443,264
model.language_model.layers.5.linear_attn.in_proj_z	16,777,216	3,168,220,480
model.language_model.layers.5.linear_attn.norm	128	3,168,220,608
model.language_model.layers.5.linear_attn.out_proj	16,777,216	3,184,997,824
model.language_model.layers.5.mlp.down_proj	50,331,648	3,235,329,472
model.language_model.layers.5.mlp.gate_proj	50,331,648	3,285,661,120
model.language_model.layers.5.mlp.up_proj	50,331,648	3,335,992,768
model.language_model.layers.5.post_attention_layernorm	4,096	3,335,996,864
model.language_model.layers.6.input_layernorm	4,096	3,336,000,960
model.language_model.layers.6.linear_attn	64	3,336,001,024
model.language_model.layers.6.linear_attn.conv1d	32,768	3,336,033,792
model.language_model.layers.6.linear_attn.in_proj_a	131,072	3,336,164,864
model.language_model.layers.6.linear_attn.in_proj_b	131,072	3,336,295,936
model.language_model.layers.6.linear_attn.in_proj_qkv	33,554,432	3,369,850,368
model.language_model.layers.6.linear_attn.in_proj_z	16,777,216	3,386,627,584
model.language_model.layers.6.linear_attn.norm	128	3,386,627,712
model.language_model.layers.6.linear_attn.out_proj	16,777,216	3,403,404,928
model.language_model.layers.6.mlp.down_proj	50,331,648	3,453,736,576
model.language_model.layers.6.mlp.gate_proj	50,331,648	3,504,068,224
model.language_model.layers.6.mlp.up_proj	50,331,648	3,554,399,872
model.language_model.layers.6.post_attention_layernorm	4,096	3,554,403,968
model.language_model.layers.7.input_layernorm	4,096	3,554,408,064
model.language_model.layers.7.mlp.down_proj	50,331,648	3,604,739,712
model.language_model.layers.7.mlp.gate_proj	50,331,648	3,655,071,360
model.language_model.layers.7.mlp.up_proj	50,331,648	3,705,403,008
model.language_model.layers.7.post_attention_layernorm	4,096	3,705,407,104
model.language_model.layers.7.self_attn.k_norm	256	3,705,407,360
model.language_model.layers.7.self_attn.k_proj	4,194,304	3,709,601,664
model.language_model.layers.7.self_attn.o_proj	16,777,216	3,726,378,880
model.language_model.layers.7.self_attn.q_norm	256	3,726,379,136
model.language_model.layers.7.self_attn.q_proj	33,554,432	3,759,933,568
model.language_model.layers.7.self_attn.v_proj	4,194,304	3,764,127,872
model.language_model.layers.8.input_layernorm	4,096	3,764,131,968
model.language_model.layers.8.linear_attn	64	3,764,132,032
model.language_model.layers.8.linear_attn.conv1d	32,768	3,764,164,800
model.language_model.layers.8.linear_attn.in_proj_a	131,072	3,764,295,872
model.language_model.layers.8.linear_attn.in_proj_b	131,072	3,764,426,944
model.language_model.layers.8.linear_attn.in_proj_qkv	33,554,432	3,797,981,376
model.language_model.layers.8.linear_attn.in_proj_z	16,777,216	3,814,758,592
model.language_model.layers.8.linear_attn.norm	128	3,814,758,720
model.language_model.layers.8.linear_attn.out_proj	16,777,216	3,831,535,936
model.language_model.layers.8.mlp.down_proj	50,331,648	3,881,867,584
model.language_model.layers.8.mlp.gate_proj	50,331,648	3,932,199,232
model.language_model.layers.8.mlp.up_proj	50,331,648	3,982,530,880

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Module	Params	Running total
model.language_model.layers.8.post_attention_layernorm	4,096	3,982,534,976
model.language_model.layers.9.input_layernorm	4,096	3,982,539,072
model.language_model.layers.9.linear_attn	64	3,982,539,136
model.language_model.layers.9.linear_attn.conv1d	32,768	3,982,571,904
model.language_model.layers.9.linear_attn.in_proj_a	131,072	3,982,702,976
model.language_model.layers.9.linear_attn.in_proj_b	131,072	3,982,834,048
model.language_model.layers.9.linear_attn.in_proj_qkv	33,554,432	4,016,388,480
model.language_model.layers.9.linear_attn.in_proj_z	16,777,216	4,033,165,696
model.language_model.layers.9.linear_attn.norm	128	4,033,165,824
model.language_model.layers.9.linear_attn.out_proj	16,777,216	4,049,943,040
model.language_model.layers.9.mlp.down_proj	50,331,648	4,100,274,688
model.language_model.layers.9.mlp.gate_proj	50,331,648	4,150,606,336
model.language_model.layers.9.mlp.up_proj	50,331,648	4,200,937,984
model.language_model.layers.9.post_attention_layernorm	4,096	4,200,942,080
model.language_model.layers.10.input_layernorm	4,096	4,200,946,176
model.language_model.layers.10.linear_attn	64	4,200,946,240
model.language_model.layers.10.linear_attn.conv1d	32,768	4,200,979,008
model.language_model.layers.10.linear_attn.in_proj_a	131,072	4,201,110,080
model.language_model.layers.10.linear_attn.in_proj_b	131,072	4,201,241,152
model.language_model.layers.10.linear_attn.in_proj_qkv	33,554,432	4,234,795,584
model.language_model.layers.10.linear_attn.in_proj_z	16,777,216	4,251,572,800
model.language_model.layers.10.linear_attn.norm	128	4,251,572,928
model.language_model.layers.10.linear_attn.out_proj	16,777,216	4,268,350,144
model.language_model.layers.10.mlp.down_proj	50,331,648	4,318,681,792
model.language_model.layers.10.mlp.gate_proj	50,331,648	4,369,013,440
model.language_model.layers.10.mlp.up_proj	50,331,648	4,419,345,088
model.language_model.layers.10.post_attention_layernorm	4,096	4,419,349,184
model.language_model.layers.11.input_layernorm	4,096	4,419,353,280
model.language_model.layers.11.mlp.down_proj	50,331,648	4,469,684,928
model.language_model.layers.11.mlp.gate_proj	50,331,648	4,520,016,576
model.language_model.layers.11.mlp.up_proj	50,331,648	4,570,348,224
model.language_model.layers.11.post_attention_layernorm	4,096	4,570,352,320
model.language_model.layers.11.self_attn.k_norm	256	4,570,352,576
model.language_model.layers.11.self_attn.k_proj	4,194,304	4,574,546,880
model.language_model.layers.11.self_attn.o_proj	16,777,216	4,591,324,096
model.language_model.layers.11.self_attn.q_norm	256	4,591,324,352
model.language_model.layers.11.self_attn.q_proj	33,554,432	4,624,878,784
model.language_model.layers.11.self_attn.v_proj	4,194,304	4,629,073,088
model.language_model.layers.12.input_layernorm	4,096	4,629,077,184
model.language_model.layers.12.linear_attn	64	4,629,077,248
model.language_model.layers.12.linear_attn.conv1d	32,768	4,629,110,016
model.language_model.layers.12.linear_attn.in_proj_a	131,072	4,629,241,088
model.language_model.layers.12.linear_attn.in_proj_b	131,072	4,629,372,160
model.language_model.layers.12.linear_attn.in_proj_qkv	33,554,432	4,662,926,592
model.language_model.layers.12.linear_attn.in_proj_z	16,777,216	4,679,703,808

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Module	Params	Running total
model.language_model.layers.12.linear_attn.norm	128	4,679,703,936
model.language_model.layers.12.linear_attn.out_proj	16,777,216	4,696,481,152
model.language_model.layers.12.mlp.down_proj	50,331,648	4,746,812,800
model.language_model.layers.12.mlp.gate_proj	50,331,648	4,797,144,448
model.language_model.layers.12.mlp.up_proj	50,331,648	4,847,476,096
model.language_model.layers.12. post_attention_layernorm	4,096	4,847,480,192
model.language_model.layers.13.input_layernorm	4,096	4,847,484,288
model.language_model.layers.13.linear_attn	64	4,847,484,352
model.language_model.layers.13.linear_attn.conv1d	32,768	4,847,517,120
model.language_model.layers.13.linear_attn. in_proj_a	131,072	4,847,648,192
model.language_model.layers.13.linear_attn. in_proj_b	131,072	4,847,779,264
model.language_model.layers.13.linear_attn. in_proj_qkv	33,554,432	4,881,333,696
model.language_model.layers.13.linear_attn. in_proj_z	16,777,216	4,898,110,912
model.language_model.layers.13.linear_attn.norm	128	4,898,111,040
model.language_model.layers.13.linear_attn.out_proj	16,777,216	4,914,888,256
model.language_model.layers.13.mlp.down_proj	50,331,648	4,965,219,904
model.language_model.layers.13.mlp.gate_proj	50,331,648	5,015,551,552
model.language_model.layers.13.mlp.up_proj	50,331,648	5,065,883,200
model.language_model.layers.13. post_attention_layernorm	4,096	5,065,887,296
model.language_model.layers.14.input_layernorm	4,096	5,065,891,392
model.language_model.layers.14.linear_attn	64	5,065,891,456
model.language_model.layers.14.linear_attn.conv1d	32,768	5,065,924,224
model.language_model.layers.14.linear_attn. in_proj_a	131,072	5,066,055,296
model.language_model.layers.14.linear_attn. in_proj_b	131,072	5,066,186,368
model.language_model.layers.14.linear_attn. in_proj_qkv	33,554,432	5,099,740,800
model.language_model.layers.14.linear_attn. in_proj_z	16,777,216	5,116,518,016
model.language_model.layers.14.linear_attn.norm	128	5,116,518,144
model.language_model.layers.14.linear_attn.out_proj	16,777,216	5,133,295,360
model.language_model.layers.14.mlp.down_proj	50,331,648	5,183,627,008
model.language_model.layers.14.mlp.gate_proj	50,331,648	5,233,958,656
model.language_model.layers.14.mlp.up_proj	50,331,648	5,284,290,304
model.language_model.layers.14. post_attention_layernorm	4,096	5,284,294,400
model.language_model.layers.15.input_layernorm	4,096	5,284,298,496
model.language_model.layers.15.mlp.down_proj	50,331,648	5,334,630,144
model.language_model.layers.15.mlp.gate_proj	50,331,648	5,384,961,792
model.language_model.layers.15.mlp.up_proj	50,331,648	5,435,293,440
model.language_model.layers.15. post_attention_layernorm	4,096	5,435,297,536
model.language_model.layers.15.self_attn.k_norm	256	5,435,297,792
model.language_model.layers.15.self_attn.k_proj	4,194,304	5,439,492,096
model.language_model.layers.15.self_attn.o_proj	16,777,216	5,456,269,312
model.language_model.layers.15.self_attn.q_norm	256	5,456,269,568
model.language_model.layers.15.self_attn.q_proj	33,554,432	5,489,824,000
model.language_model.layers.15.self_attn.v_proj	4,194,304	5,494,018,304
model.language_model.layers.16.input_layernorm	4,096	5,494,022,400
model.language_model.layers.16.linear_attn	64	5,494,022,464
model.language_model.layers.16.linear_attn.conv1d	32,768	5,494,055,232

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Module	Params	Running total
model.language_model.layers.16.linear_attn.in_proj_a	131,072	5,494,186,304
model.language_model.layers.16.linear_attn.in_proj_b	131,072	5,494,317,376
model.language_model.layers.16.linear_attn.in_proj_qkv	33,554,432	5,527,871,808
model.language_model.layers.16.linear_attn.in_proj_z	16,777,216	5,544,649,024
model.language_model.layers.16.linear_attn.norm	128	5,544,649,152
model.language_model.layers.16.linear_attn.out_proj	16,777,216	5,561,426,368
model.language_model.layers.16.mlp.down_proj	50,331,648	5,611,758,016
model.language_model.layers.16.mlp.gate_proj	50,331,648	5,662,089,664
model.language_model.layers.16.mlp.up_proj	50,331,648	5,712,421,312
model.language_model.layers.16.post_attention_layernorm	4,096	5,712,425,408
model.language_model.layers.17.input_layernorm	4,096	5,712,429,504
model.language_model.layers.17.linear_attn	64	5,712,429,568
model.language_model.layers.17.linear_attn.conv1d	32,768	5,712,462,336
model.language_model.layers.17.linear_attn.in_proj_a	131,072	5,712,593,408
model.language_model.layers.17.linear_attn.in_proj_b	131,072	5,712,724,480
model.language_model.layers.17.linear_attn.in_proj_qkv	33,554,432	5,746,278,912
model.language_model.layers.17.linear_attn.in_proj_z	16,777,216	5,763,056,128
model.language_model.layers.17.linear_attn.norm	128	5,763,056,256
model.language_model.layers.17.linear_attn.out_proj	16,777,216	5,779,833,472
model.language_model.layers.17.mlp.down_proj	50,331,648	5,830,165,120
model.language_model.layers.17.mlp.gate_proj	50,331,648	5,880,496,768
model.language_model.layers.17.mlp.up_proj	50,331,648	5,930,828,416
model.language_model.layers.17.post_attention_layernorm	4,096	5,930,832,512
model.language_model.layers.18.input_layernorm	4,096	5,930,836,608
model.language_model.layers.18.linear_attn	64	5,930,836,672
model.language_model.layers.18.linear_attn.conv1d	32,768	5,930,869,440
model.language_model.layers.18.linear_attn.in_proj_a	131,072	5,931,000,512
model.language_model.layers.18.linear_attn.in_proj_b	131,072	5,931,131,584
model.language_model.layers.18.linear_attn.in_proj_qkv	33,554,432	5,964,686,016
model.language_model.layers.18.linear_attn.in_proj_z	16,777,216	5,981,463,232
model.language_model.layers.18.linear_attn.norm	128	5,981,463,360
model.language_model.layers.18.linear_attn.out_proj	16,777,216	5,998,240,576
model.language_model.layers.18.mlp.down_proj	50,331,648	6,048,572,224
model.language_model.layers.18.mlp.gate_proj	50,331,648	6,098,903,872
model.language_model.layers.18.mlp.up_proj	50,331,648	6,149,235,520
model.language_model.layers.18.post_attention_layernorm	4,096	6,149,239,616
model.language_model.layers.19.input_layernorm	4,096	6,149,243,712
model.language_model.layers.19.mlp.down_proj	50,331,648	6,199,575,360
model.language_model.layers.19.mlp.gate_proj	50,331,648	6,249,907,008
model.language_model.layers.19.mlp.up_proj	50,331,648	6,300,238,656
model.language_model.layers.19.post_attention_layernorm	4,096	6,300,242,752
model.language_model.layers.19.self_attn.k_norm	256	6,300,243,008

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Module	Params	Running total
model.language_model.layers.19.self_attn.k_proj	4,194,304	6,304,437,312
model.language_model.layers.19.self_attn.o_proj	16,777,216	6,321,214,528
model.language_model.layers.19.self_attn.q_norm	256	6,321,214,784
model.language_model.layers.19.self_attn.q_proj	33,554,432	6,354,769,216
model.language_model.layers.19.self_attn.v_proj	4,194,304	6,358,963,520
model.language_model.layers.20.input_layernorm	4,096	6,358,967,616
model.language_model.layers.20.linear_attn	64	6,358,967,680
model.language_model.layers.20.linear_attn.conv1d	32,768	6,359,000,448
model.language_model.layers.20.linear_attn.in_proj_a	131,072	6,359,131,520
model.language_model.layers.20.linear_attn.in_proj_b	131,072	6,359,262,592
model.language_model.layers.20.linear_attn.in_proj_qkv	33,554,432	6,392,817,024
model.language_model.layers.20.linear_attn.in_proj_z	16,777,216	6,409,594,240
model.language_model.layers.20.linear_attn.norm	128	6,409,594,368
model.language_model.layers.20.linear_attn.out_proj	16,777,216	6,426,371,584
model.language_model.layers.20.mlp.down_proj	50,331,648	6,476,703,232
model.language_model.layers.20.mlp.gate_proj	50,331,648	6,527,034,880
model.language_model.layers.20.mlp.up_proj	50,331,648	6,577,366,528
model.language_model.layers.20.post_attention_layernorm	4,096	6,577,370,624
model.language_model.layers.21.input_layernorm	4,096	6,577,374,720
model.language_model.layers.21.linear_attn	64	6,577,374,784
model.language_model.layers.21.linear_attn.conv1d	32,768	6,577,407,552
model.language_model.layers.21.linear_attn.in_proj_a	131,072	6,577,538,624
model.language_model.layers.21.linear_attn.in_proj_b	131,072	6,577,669,696
model.language_model.layers.21.linear_attn.in_proj_qkv	33,554,432	6,611,224,128
model.language_model.layers.21.linear_attn.in_proj_z	16,777,216	6,628,001,344
model.language_model.layers.21.linear_attn.norm	128	6,628,001,472
model.language_model.layers.21.linear_attn.out_proj	16,777,216	6,644,778,688
model.language_model.layers.21.mlp.down_proj	50,331,648	6,695,110,336
model.language_model.layers.21.mlp.gate_proj	50,331,648	6,745,441,984
model.language_model.layers.21.mlp.up_proj	50,331,648	6,795,773,632
model.language_model.layers.21.post_attention_layernorm	4,096	6,795,777,728
model.language_model.layers.22.input_layernorm	4,096	6,795,781,824
model.language_model.layers.22.linear_attn	64	6,795,781,888
model.language_model.layers.22.linear_attn.conv1d	32,768	6,795,814,656
model.language_model.layers.22.linear_attn.in_proj_a	131,072	6,795,945,728
model.language_model.layers.22.linear_attn.in_proj_b	131,072	6,796,076,800
model.language_model.layers.22.linear_attn.in_proj_qkv	33,554,432	6,829,631,232
model.language_model.layers.22.linear_attn.in_proj_z	16,777,216	6,846,408,448
model.language_model.layers.22.linear_attn.norm	128	6,846,408,576
model.language_model.layers.22.linear_attn.out_proj	16,777,216	6,863,185,792
model.language_model.layers.22.mlp.down_proj	50,331,648	6,913,517,440
model.language_model.layers.22.mlp.gate_proj	50,331,648	6,963,849,088
model.language_model.layers.22.mlp.up_proj	50,331,648	7,014,180,736

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Module	Params	Running total
model.language_model.layers.22.post_attention_layernorm	4,096	7,014,184,832
model.language_model.layers.23.input_layernorm	4,096	7,014,188,928
model.language_model.layers.23.mlp.down_proj	50,331,648	7,064,520,576
model.language_model.layers.23.mlp.gate_proj	50,331,648	7,114,852,224
model.language_model.layers.23.mlp.up_proj	50,331,648	7,165,183,872
model.language_model.layers.23.post_attention_layernorm	4,096	7,165,187,968
model.language_model.layers.23.self_attn.k_norm	256	7,165,188,224
model.language_model.layers.23.self_attn.k_proj	4,194,304	7,169,382,528
model.language_model.layers.23.self_attn.o_proj	16,777,216	7,186,159,744
model.language_model.layers.23.self_attn.q_norm	256	7,186,160,000
model.language_model.layers.23.self_attn.q_proj	33,554,432	7,219,714,432
model.language_model.layers.23.self_attn.v_proj	4,194,304	7,223,908,736
model.language_model.layers.24.input_layernorm	4,096	7,223,912,832
model.language_model.layers.24.linear_attn	64	7,223,912,896
model.language_model.layers.24.linear_attn.conv1d	32,768	7,223,945,664
model.language_model.layers.24.linear_attn.in_proj_a	131,072	7,224,076,736
model.language_model.layers.24.linear_attn.in_proj_b	131,072	7,224,207,808
model.language_model.layers.24.linear_attn.in_proj_qkv	33,554,432	7,257,762,240
model.language_model.layers.24.linear_attn.in_proj_z	16,777,216	7,274,539,456
model.language_model.layers.24.linear_attn.norm	128	7,274,539,584
model.language_model.layers.24.linear_attn.out_proj	16,777,216	7,291,316,800
model.language_model.layers.24.mlp.down_proj	50,331,648	7,341,648,448
model.language_model.layers.24.mlp.gate_proj	50,331,648	7,391,980,096
model.language_model.layers.24.mlp.up_proj	50,331,648	7,442,311,744
model.language_model.layers.24.post_attention_layernorm	4,096	7,442,315,840
model.language_model.layers.25.input_layernorm	4,096	7,442,319,936
model.language_model.layers.25.linear_attn	64	7,442,320,000
model.language_model.layers.25.linear_attn.conv1d	32,768	7,442,352,768
model.language_model.layers.25.linear_attn.in_proj_a	131,072	7,442,483,840
model.language_model.layers.25.linear_attn.in_proj_b	131,072	7,442,614,912
model.language_model.layers.25.linear_attn.in_proj_qkv	33,554,432	7,476,169,344
model.language_model.layers.25.linear_attn.in_proj_z	16,777,216	7,492,946,560
model.language_model.layers.25.linear_attn.norm	128	7,492,946,688
model.language_model.layers.25.linear_attn.out_proj	16,777,216	7,509,723,904
model.language_model.layers.25.mlp.down_proj	50,331,648	7,560,055,552
model.language_model.layers.25.mlp.gate_proj	50,331,648	7,610,387,200
model.language_model.layers.25.mlp.up_proj	50,331,648	7,660,718,848
model.language_model.layers.25.post_attention_layernorm	4,096	7,660,722,944
model.language_model.layers.26.input_layernorm	4,096	7,660,727,040
model.language_model.layers.26.linear_attn	64	7,660,727,104
model.language_model.layers.26.linear_attn.conv1d	32,768	7,660,759,872
model.language_model.layers.26.linear_attn.in_proj_a	131,072	7,660,890,944
model.language_model.layers.26.linear_attn.in_proj_b	131,072	7,661,022,016

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Module	Params	Running total
model.language_model.layers.26.linear_attn.in_proj_qkv	33,554,432	7,694,576,448
model.language_model.layers.26.linear_attn.in_proj_z	16,777,216	7,711,353,664
model.language_model.layers.26.linear_attn.norm	128	7,711,353,792
model.language_model.layers.26.linear_attn.out_proj	16,777,216	7,728,131,008
model.language_model.layers.26.mlp.down_proj	50,331,648	7,778,462,656
model.language_model.layers.26.mlp.gate_proj	50,331,648	7,828,794,304
model.language_model.layers.26.mlp.up_proj	50,331,648	7,879,125,952
model.language_model.layers.26.post_attention_layernorm	4,096	7,879,130,048
model.language_model.layers.27.input_layernorm	4,096	7,879,134,144
model.language_model.layers.27.mlp.down_proj	50,331,648	7,929,465,792
model.language_model.layers.27.mlp.gate_proj	50,331,648	7,979,797,440
model.language_model.layers.27.mlp.up_proj	50,331,648	8,030,129,088
model.language_model.layers.27.post_attention_layernorm	4,096	8,030,133,184
model.language_model.layers.27.self_attn.k_norm	256	8,030,133,440
model.language_model.layers.27.self_attn.k_proj	4,194,304	8,034,327,744
model.language_model.layers.27.self_attn.o_proj	16,777,216	8,051,104,960
model.language_model.layers.27.self_attn.q_norm	256	8,051,105,216
model.language_model.layers.27.self_attn.q_proj	33,554,432	8,084,659,648
model.language_model.layers.27.self_attn.v_proj	4,194,304	8,088,853,952
model.language_model.layers.28.input_layernorm	4,096	8,088,858,048
model.language_model.layers.28.linear_attn	64	8,088,858,112
model.language_model.layers.28.linear_attn.conv1d	32,768	8,088,890,880
model.language_model.layers.28.linear_attn.in_proj_a	131,072	8,089,021,952
model.language_model.layers.28.linear_attn.in_proj_b	131,072	8,089,153,024
model.language_model.layers.28.linear_attn.in_proj_qkv	33,554,432	8,122,707,456
model.language_model.layers.28.linear_attn.in_proj_z	16,777,216	8,139,484,672
model.language_model.layers.28.linear_attn.norm	128	8,139,484,800
model.language_model.layers.28.linear_attn.out_proj	16,777,216	8,156,262,016
model.language_model.layers.28.mlp.down_proj	50,331,648	8,206,593,664
model.language_model.layers.28.mlp.gate_proj	50,331,648	8,256,925,312
model.language_model.layers.28.mlp.up_proj	50,331,648	8,307,256,960
model.language_model.layers.28.post_attention_layernorm	4,096	8,307,261,056
model.language_model.layers.29.input_layernorm	4,096	8,307,265,152
model.language_model.layers.29.linear_attn	64	8,307,265,216
model.language_model.layers.29.linear_attn.conv1d	32,768	8,307,297,984
model.language_model.layers.29.linear_attn.in_proj_a	131,072	8,307,429,056
model.language_model.layers.29.linear_attn.in_proj_b	131,072	8,307,560,128
model.language_model.layers.29.linear_attn.in_proj_qkv	33,554,432	8,341,114,560
model.language_model.layers.29.linear_attn.in_proj_z	16,777,216	8,357,891,776
model.language_model.layers.29.linear_attn.norm	128	8,357,891,904
model.language_model.layers.29.linear_attn.out_proj	16,777,216	8,374,669,120
model.language_model.layers.29.mlp.down_proj	50,331,648	8,425,000,768
model.language_model.layers.29.mlp.gate_proj	50,331,648	8,475,332,416
model.language_model.layers.29.mlp.up_proj	50,331,648	8,525,664,064

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Module	Params	Running total
model.language_model.layers.29. post_attention_layernorm	4,096	8,525,668,160
model.language_model.layers.30.input_layernorm	4,096	8,525,672,256
model.language_model.layers.30.linear_attn	64	8,525,672,320
model.language_model.layers.30.linear_attn.conv1d	32,768	8,525,705,088
model.language_model.layers.30.linear_attn. in_proj_a	131,072	8,525,836,160
model.language_model.layers.30.linear_attn. in_proj_b	131,072	8,525,967,232
model.language_model.layers.30.linear_attn. in_proj_qkv	33,554,432	8,559,521,664
model.language_model.layers.30.linear_attn. in_proj_z	16,777,216	8,576,298,880
model.language_model.layers.30.linear_attn.norm	128	8,576,299,008
model.language_model.layers.30.linear_attn.out_proj	16,777,216	8,593,076,224
model.language_model.layers.30.mlp.down_proj	50,331,648	8,643,407,872
model.language_model.layers.30.mlp.gate_proj	50,331,648	8,693,739,520
model.language_model.layers.30.mlp.up_proj	50,331,648	8,744,071,168
model.language_model.layers.30. post_attention_layernorm	4,096	8,744,075,264
model.language_model.layers.31.input_layernorm	4,096	8,744,079,360
model.language_model.layers.31.mlp.down_proj	50,331,648	8,794,411,008
model.language_model.layers.31.mlp.gate_proj	50,331,648	8,844,742,656
model.language_model.layers.31.mlp.up_proj	50,331,648	8,895,074,304
model.language_model.layers.31. post_attention_layernorm	4,096	8,895,078,400
model.language_model.layers.31.self_attn.k_norm	256	8,895,078,656
model.language_model.layers.31.self_attn.k_proj	4,194,304	8,899,272,960
model.language_model.layers.31.self_attn.o_proj	16,777,216	8,916,050,176
model.language_model.layers.31.self_attn.q_norm	256	8,916,050,432
model.language_model.layers.31.self_attn.q_proj	33,554,432	8,949,604,864
model.language_model.layers.31.self_attn.v_proj	4,194,304	8,953,799,168
model.language_model.norm	4,096	8,953,803,264
model.visual.blocks.0.attn.proj	1,328,256	8,955,131,520
model.visual.blocks.0.attn.qkv	3,984,768	8,959,116,288
model.visual.blocks.0.mlp.linear_fc1	4,962,512	8,964,078,800
model.visual.blocks.0.mlp.linear_fc2	4,959,360	8,969,038,160
model.visual.blocks.0.norm1	2,304	8,969,040,464
model.visual.blocks.0.norm2	2,304	8,969,042,768
model.visual.blocks.1.attn.proj	1,328,256	8,970,371,024
model.visual.blocks.1.attn.qkv	3,984,768	8,974,355,792
model.visual.blocks.1.mlp.linear_fc1	4,962,512	8,979,318,304
model.visual.blocks.1.mlp.linear_fc2	4,959,360	8,984,277,664
model.visual.blocks.1.norm1	2,304	8,984,279,968
model.visual.blocks.1.norm2	2,304	8,984,282,272
model.visual.blocks.2.attn.proj	1,328,256	8,985,610,528
model.visual.blocks.2.attn.qkv	3,984,768	8,989,595,296
model.visual.blocks.2.mlp.linear_fc1	4,962,512	8,994,557,808
model.visual.blocks.2.mlp.linear_fc2	4,959,360	8,999,517,168
model.visual.blocks.2.norm1	2,304	8,999,519,472
model.visual.blocks.2.norm2	2,304	8,999,521,776
model.visual.blocks.3.attn.proj	1,328,256	9,000,850,032
model.visual.blocks.3.attn.qkv	3,984,768	9,004,834,800
model.visual.blocks.3.mlp.linear_fc1	4,962,512	9,009,797,312
model.visual.blocks.3.mlp.linear_fc2	4,959,360	9,014,756,672
model.visual.blocks.3.norm1	2,304	9,014,758,976
model.visual.blocks.3.norm2	2,304	9,014,761,280
model.visual.blocks.4.attn.proj	1,328,256	9,016,089,536

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Module	Params	Running total
model.visual.blocks.4.attn.qkv	3,984,768	9,020,074,304
model.visual.blocks.4.mlp.linear_fc1	4,962,512	9,025,036,816
model.visual.blocks.4.mlp.linear_fc2	4,959,360	9,029,996,176
model.visual.blocks.4.norm1	2,304	9,029,998,480
model.visual.blocks.4.norm2	2,304	9,030,000,784
model.visual.blocks.5.attn.proj	1,328,256	9,031,329,040
model.visual.blocks.5.attn.qkv	3,984,768	9,035,313,808
model.visual.blocks.5.mlp.linear_fc1	4,962,512	9,040,276,320
model.visual.blocks.5.mlp.linear_fc2	4,959,360	9,045,235,680
model.visual.blocks.5.norm1	2,304	9,045,237,984
model.visual.blocks.5.norm2	2,304	9,045,240,288
model.visual.blocks.6.attn.proj	1,328,256	9,046,568,544
model.visual.blocks.6.attn.qkv	3,984,768	9,050,553,312
model.visual.blocks.6.mlp.linear_fc1	4,962,512	9,055,515,824
model.visual.blocks.6.mlp.linear_fc2	4,959,360	9,060,475,184
model.visual.blocks.6.norm1	2,304	9,060,477,488
model.visual.blocks.6.norm2	2,304	9,060,479,792
model.visual.blocks.7.attn.proj	1,328,256	9,061,808,048
model.visual.blocks.7.attn.qkv	3,984,768	9,065,792,816
model.visual.blocks.7.mlp.linear_fc1	4,962,512	9,070,755,328
model.visual.blocks.7.mlp.linear_fc2	4,959,360	9,075,714,688
model.visual.blocks.7.norm1	2,304	9,075,716,992
model.visual.blocks.7.norm2	2,304	9,075,719,296
model.visual.blocks.8.attn.proj	1,328,256	9,077,047,552
model.visual.blocks.8.attn.qkv	3,984,768	9,081,032,320
model.visual.blocks.8.mlp.linear_fc1	4,962,512	9,085,994,832
model.visual.blocks.8.mlp.linear_fc2	4,959,360	9,090,954,192
model.visual.blocks.8.norm1	2,304	9,090,956,496
model.visual.blocks.8.norm2	2,304	9,090,958,800
model.visual.blocks.9.attn.proj	1,328,256	9,092,287,056
model.visual.blocks.9.attn.qkv	3,984,768	9,096,271,824
model.visual.blocks.9.mlp.linear_fc1	4,962,512	9,101,234,336
model.visual.blocks.9.mlp.linear_fc2	4,959,360	9,106,193,696
model.visual.blocks.9.norm1	2,304	9,106,196,000
model.visual.blocks.9.norm2	2,304	9,106,198,304
model.visual.blocks.10.attn.proj	1,328,256	9,107,526,560
model.visual.blocks.10.attn.qkv	3,984,768	9,111,511,328
model.visual.blocks.10.mlp.linear_fc1	4,962,512	9,116,473,840
model.visual.blocks.10.mlp.linear_fc2	4,959,360	9,121,433,200
model.visual.blocks.10.norm1	2,304	9,121,435,504
model.visual.blocks.10.norm2	2,304	9,121,437,808
model.visual.blocks.11.attn.proj	1,328,256	9,122,766,064
model.visual.blocks.11.attn.qkv	3,984,768	9,126,750,832
model.visual.blocks.11.mlp.linear_fc1	4,962,512	9,131,713,344
model.visual.blocks.11.mlp.linear_fc2	4,959,360	9,136,672,704
model.visual.blocks.11.norm1	2,304	9,136,675,008
model.visual.blocks.11.norm2	2,304	9,136,677,312
model.visual.blocks.12.attn.proj	1,328,256	9,138,005,568
model.visual.blocks.12.attn.qkv	3,984,768	9,141,990,336
model.visual.blocks.12.mlp.linear_fc1	4,962,512	9,146,952,848
model.visual.blocks.12.mlp.linear_fc2	4,959,360	9,151,912,208
model.visual.blocks.12.norm1	2,304	9,151,914,512
model.visual.blocks.12.norm2	2,304	9,151,916,816
model.visual.blocks.13.attn.proj	1,328,256	9,153,245,072
model.visual.blocks.13.attn.qkv	3,984,768	9,157,229,840
model.visual.blocks.13.mlp.linear_fc1	4,962,512	9,162,192,352
model.visual.blocks.13.mlp.linear_fc2	4,959,360	9,167,151,712
model.visual.blocks.13.norm1	2,304	9,167,154,016

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Module	Params	Running total
model.visual.blocks.13.norm2	2,304	9,167,156,320
model.visual.blocks.14.attn.proj	1,328,256	9,168,484,576
model.visual.blocks.14.attn.qkv	3,984,768	9,172,469,344
model.visual.blocks.14.mlp.linear_fc1	4,962,512	9,177,431,856
model.visual.blocks.14.mlp.linear_fc2	4,959,360	9,182,391,216
model.visual.blocks.14.norm1	2,304	9,182,393,520
model.visual.blocks.14.norm2	2,304	9,182,395,824
model.visual.blocks.15.attn.proj	1,328,256	9,183,724,080
model.visual.blocks.15.attn.qkv	3,984,768	9,187,708,848
model.visual.blocks.15.mlp.linear_fc1	4,962,512	9,192,671,360
model.visual.blocks.15.mlp.linear_fc2	4,959,360	9,197,630,720
model.visual.blocks.15.norm1	2,304	9,197,633,024
model.visual.blocks.15.norm2	2,304	9,197,635,328
model.visual.blocks.16.attn.proj	1,328,256	9,198,963,584
model.visual.blocks.16.attn.qkv	3,984,768	9,202,948,352
model.visual.blocks.16.mlp.linear_fc1	4,962,512	9,207,910,864
model.visual.blocks.16.mlp.linear_fc2	4,959,360	9,212,870,224
model.visual.blocks.16.norm1	2,304	9,212,872,528
model.visual.blocks.16.norm2	2,304	9,212,874,832
model.visual.blocks.17.attn.proj	1,328,256	9,214,203,088
model.visual.blocks.17.attn.qkv	3,984,768	9,218,187,856
model.visual.blocks.17.mlp.linear_fc1	4,962,512	9,223,150,368
model.visual.blocks.17.mlp.linear_fc2	4,959,360	9,228,109,728
model.visual.blocks.17.norm1	2,304	9,228,112,032
model.visual.blocks.17.norm2	2,304	9,228,114,336
model.visual.blocks.18.attn.proj	1,328,256	9,229,442,592
model.visual.blocks.18.attn.qkv	3,984,768	9,233,427,360
model.visual.blocks.18.mlp.linear_fc1	4,962,512	9,238,389,872
model.visual.blocks.18.mlp.linear_fc2	4,959,360	9,243,349,232
model.visual.blocks.18.norm1	2,304	9,243,351,536
model.visual.blocks.18.norm2	2,304	9,243,353,840
model.visual.blocks.19.attn.proj	1,328,256	9,244,682,096
model.visual.blocks.19.attn.qkv	3,984,768	9,248,666,864
model.visual.blocks.19.mlp.linear_fc1	4,962,512	9,253,629,376
model.visual.blocks.19.mlp.linear_fc2	4,959,360	9,258,588,736
model.visual.blocks.19.norm1	2,304	9,258,591,040
model.visual.blocks.19.norm2	2,304	9,258,593,344
model.visual.blocks.20.attn.proj	1,328,256	9,259,921,600
model.visual.blocks.20.attn.qkv	3,984,768	9,263,906,368
model.visual.blocks.20.mlp.linear_fc1	4,962,512	9,268,868,880
model.visual.blocks.20.mlp.linear_fc2	4,959,360	9,273,828,240
model.visual.blocks.20.norm1	2,304	9,273,830,544
model.visual.blocks.20.norm2	2,304	9,273,832,848
model.visual.blocks.21.attn.proj	1,328,256	9,275,161,104
model.visual.blocks.21.attn.qkv	3,984,768	9,279,145,872
model.visual.blocks.21.mlp.linear_fc1	4,962,512	9,284,108,384
model.visual.blocks.21.mlp.linear_fc2	4,959,360	9,289,067,744
model.visual.blocks.21.norm1	2,304	9,289,070,048
model.visual.blocks.21.norm2	2,304	9,289,072,352
model.visual.blocks.22.attn.proj	1,328,256	9,290,400,608
model.visual.blocks.22.attn.qkv	3,984,768	9,294,385,376
model.visual.blocks.22.mlp.linear_fc1	4,962,512	9,299,347,888
model.visual.blocks.22.mlp.linear_fc2	4,959,360	9,304,307,248
model.visual.blocks.22.norm1	2,304	9,304,309,552
model.visual.blocks.22.norm2	2,304	9,304,311,856
model.visual.blocks.23.attn.proj	1,328,256	9,305,640,112
model.visual.blocks.23.attn.qkv	3,984,768	9,309,624,880
model.visual.blocks.23.mlp.linear_fc1	4,962,512	9,314,587,392

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Module	Params	Running total
model.visual.blocks.23.mlp.linear_fc2	4,959,360	9,319,546,752
model.visual.blocks.23.norm1	2,304	9,319,549,056
model.visual.blocks.23.norm2	2,304	9,319,551,360
model.visual.blocks.24.attn.proj	1,328,256	9,320,879,616
model.visual.blocks.24.attn.qkv	3,984,768	9,324,864,384
model.visual.blocks.24.mlp.linear_fc1	4,962,512	9,329,826,896
model.visual.blocks.24.mlp.linear_fc2	4,959,360	9,334,786,256
model.visual.blocks.24.norm1	2,304	9,334,788,560
model.visual.blocks.24.norm2	2,304	9,334,790,864
model.visual.blocks.25.attn.proj	1,328,256	9,336,119,120
model.visual.blocks.25.attn.qkv	3,984,768	9,340,103,888
model.visual.blocks.25.mlp.linear_fc1	4,962,512	9,345,066,400
model.visual.blocks.25.mlp.linear_fc2	4,959,360	9,350,025,760
model.visual.blocks.25.norm1	2,304	9,350,028,064
model.visual.blocks.25.norm2	2,304	9,350,030,368
model.visual.blocks.26.attn.proj	1,328,256	9,351,358,624
model.visual.blocks.26.attn.qkv	3,984,768	9,355,343,392
model.visual.blocks.26.mlp.linear_fc1	4,962,512	9,360,305,904
model.visual.blocks.26.mlp.linear_fc2	4,959,360	9,365,265,264
model.visual.blocks.26.norm1	2,304	9,365,267,568
model.visual.blocks.26.norm2	2,304	9,365,269,872
model.visual.merger.linear_fc1	21,238,272	9,386,508,144
model.visual.merger.linear_fc2	18,878,464	9,405,386,608
model.visual.merger.norm	2,304	9,405,388,912
model.visual.patch_embed.proj	1,770,624	9,407,159,536
model.visual.pos_embed	2,654,208	9,409,813,744